

EVERY NONRECURSIVE MANY-ONE DEGREE CONTAINS EITHER ONE OR INFINITELY MANY FINITE-ONE DEGREES

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ABSTRACT. We prove that every nonrecursive many-one degree contains either exactly one or infinitely many finite-one degrees. This gives a negative answer to Open Question 2 of Richter, Stephan, and Zhang [6, p. 17]. In earlier work, the author showed that, for almost every $A \subseteq \mathbb{N}$ with respect to the standard product measure on $2^{\mathbb{N}}$, the degree $[A]_m$ contains an infinite antichain of finite-one degrees [1]. Thus the question had already received an almost-sure negative answer; the present theorem settles it for every nonrecursive many-one degree.

1. INTRODUCTION

Finite-one reducibility is the restriction of many-one reducibility in which every fibre of the reducing function is finite. Thus, for all sets X, Y ,

$$X \leq_1 Y \implies X \leq_{\text{fo}} Y \implies X \leq_m Y.$$

Given a many-one degree $\mathbf{d}$, one may therefore ask how many finite-one degrees occur inside $\mathbf{d}$, and how they are ordered.

Richter, Stephan, and Zhang asked whether there exists a nonrecursive many-one degree containing at least two but only finitely many finite-one degrees [6, p. 17]. In earlier work, the author proved that the many-one degree of every m -rigid set contains an infinite antichain of finite-one degrees [1]. Since the class of m -rigid sets has Lebesgue measure 1 and is comeager, this gave an almost-sure and comeager negative answer to the question. The present paper removes the m -rigidity hypothesis and settles the question in full: no such degree exists. More precisely, every nonrecursive many-one degree contains either exactly one or infinitely many finite-one degrees.

For every set A , the cylinder

$$\text{Cyl}(A) = \{\langle x, n \rangle : x \in A, n \in \mathbb{N}\}$$

belongs to the same many-one degree as A , and $[\text{Cyl}(A)]_{\text{fo}}$ is the greatest finite-one degree in that many-one degree. Maslova proved that every nonrecursive c.e. many-one degree contains either exactly one or infinitely many finite-one degrees [2]. We shall also use the same conclusion when the complement of a representative is c.e., by passing to the complementary many-one degree and using complementation.

The main new ingredient is the following local interpolation theorem. If A is nonrecursive, is not a cylinder, and neither A nor $\bar{A}$ is c.e., then there exists $C \equiv_m A$ such that

$$[A]_{\text{fo}} < [C]_{\text{fo}} < [\text{Cyl}(A)]_{\text{fo}}.$$

Thus, under these hypotheses, $[A]_{\text{fo}}$ can be strictly interpolated below the greatest finite-one degree.

The dichotomy follows quickly. Suppose that a nonrecursive many-one degree contained finitely many but more than one finite-one degrees. Let T be its greatest finite-one degree, and choose $E < T$ which is covered by T , that is, such that there is no finite-one degree D with

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$$E < D < T.$$

Such an E exists because there are only finitely many finite-one degrees. Choose A representing E . Then A is not a cylinder. If A is c.e., or if $\overline{A}$ is c.e., Maslova's theorem gives a contradiction, in the second case after complementation. Otherwise the local interpolation theorem produces

$$E < [C]_{\text{fo}} < T,$$

again a contradiction.

The proof of the local interpolation theorem is based on a priority construction relative to $0'$. We construct a set $N \in \Sigma_2^0$ and let

$$P = \mathbb{N} \setminus N.$$

A profile-realization lemma then yields a total computable surjection p whose infinite-fibre profile is exactly P , and we set

$$C = p^{-1}(A).$$

Two families of requirements control the two strict inequalities. The upper requirements place suitable finite rows inside N ; after profile realization, the corresponding values have finite p -fibres. The lower requirements preserve markers outside N and outside the domains of partial ascending self-reductions of A ; these markers therefore become points of P . The construction uses $0'$ only to decide fixed Π_1^0 stabilization and nonconvergence questions; in particular, it never uses $0''$ to decide whether an arbitrary c.e. set is finite.

2. PRELIMINARIES

We assume familiarity with the basic notions and notation of computability theory. Standard references are Odifreddi [3, 4], Rogers [5], and Soare [7].

We identify each set with its characteristic function. Fix a computable bijective pairing function

$$\langle \cdot, \cdot \rangle : \mathbb{N}^2 \rightarrow \mathbb{N}$$

with computable projections π_1 and π_2 .

For sets $X, Y \subseteq \mathbb{N}$, write $X \leq_m Y$ if there is a total computable function f such that

$$x \in X \iff f(x) \in Y$$

for every x . Write $X \leq_1 Y$ if such a reduction can be chosen injective, and write $X \leq_{\text{fo}} Y$ if such a reduction can be chosen so that every fibre

$$f^{-1}(y)$$

is finite.

For $\rho \in \{m, 1, \text{fo}\}$, write

$$X \equiv_\rho Y \iff X \leq_\rho Y \text{ and } Y \leq_\rho X.$$

Write

$$X <_\rho Y \iff X \leq_\rho Y \text{ and } Y \not\leq_\rho X.$$

The ρ -degree of X is the equivalence class

$$[X]_\rho = \{Y \subseteq \mathbb{N} : Y \equiv_\rho X\}.$$

The ρ -degrees are ordered by

$$[X]_\rho \leq [Y]_\rho \iff X \leq_\rho Y.$$

For the corresponding strict order, we have

$$[X]_\rho < [Y]_\rho \iff X <_\rho Y.$$

Here $\leq$ and $<$ denote the induced non-strict and strict orders on ρ -degrees, respectively.

In particular, $[X]_m$ and $[X]_{fo}$ denote the many-one degree and the finite-one degree of X , respectively.

A many-one degree is called *c.e.* if it contains a c.e. set, and *recursive* if it contains a recursive set.

For $A \subseteq \mathbb{N}$, let

$$\text{Cyl}(A) = \{\langle x, n \rangle : x \in A, n \in \mathbb{N}\}.$$

We call A a *cylinder* if

$$A \equiv_1 \text{Cyl}(A).$$

The maps

$$x \mapsto \langle x, 0 \rangle \quad \text{and} \quad \langle x, n \rangle \mapsto x$$

show that

$$A \equiv_m \text{Cyl}(A).$$

Moreover, if $B \in [A]_m$ and f is a many-one reduction of B to A , then

$$x \mapsto \langle f(x), x \rangle$$

is a one-one reduction of B to $\text{Cyl}(A)$. Hence $[\text{Cyl}(A)]_{fo}$ is the greatest finite-one degree inside $[A]_m$.

If $p : \mathbb{N} \rightarrow \mathbb{N}$ is total computable, define its infinite-fibre profile by

$$I_p = \{y : |p^{-1}(y)| = \infty\}.$$

A *section* of a map $p : \mathbb{N} \rightarrow \mathbb{N}$ is a map $s : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$p(s(y)) = y \quad \text{for every } y \in \mathbb{N}.$$

Thus a section is a right inverse of p . It is called *computable* if s is computable.

The following elementary realization lemma allows us to prescribe exactly which values of a computable surjection have infinite fibres.

Lemma 1 (Profile realization). *For every $P \in \Pi_2^0$ there is a total computable surjection $p : \mathbb{N} \rightarrow \mathbb{N}$ such that*

$$I_p = P.$$

Moreover, p has a computable section.

Proof. Choose a computable predicate Q such that

$$y \in P \iff \forall j \exists s Q(y, j, s).$$

For each $k \in \mathbb{N}$, let

$$H_k = \{y : \forall j < k \exists s Q(y, j, s)\}.$$

The sequence $(H_k)_{k \in \mathbb{N}}$ is uniformly c.e., and

$$\mathbb{N} = H_0 \supseteq H_1 \supseteq H_2 \supseteq \dots$$

Moreover,

$$P = \bigcap_{k \in \mathbb{N}} H_k.$$

Define

$$R = \{\langle 0, y \rangle : y \in \mathbb{N}\} \cup \{\langle k+1, y \rangle : k \in \mathbb{N}, y \in H_k\}.$$

Since R is an infinite c.e. set, it admits a total computable bijection

$$r : \mathbb{N} \rightarrow R,$$

obtained by enumerating R without repetitions. Define

$$p(n) = \pi_2(r(n)).$$

Then p is total computable. Since r is onto R and $\langle 0, y \rangle \in R$ for every y , the function p is surjective.

For every y ,

$$|p^{-1}(y)| = 1 + |\{k : y \in H_k\}|.$$

If $y \in P$, then $y \in H_k$ for every k , so $p^{-1}(y)$ is infinite. If $y \notin P$, then $y \notin H_k$ for all sufficiently large k , because the sequence (H_k) is decreasing. Hence $p^{-1}(y)$ is finite. Therefore

$$I_p = P.$$

Finally, define

$$s(y) = \mu n [p(n) = y].$$

The function s is total computable because p is computable and surjective, and

$$p(s(y)) = y$$

for every y . Thus s is a computable section of p .

Moreover, if $s(y) = s(z)$, then

$$y = p(s(y)) = p(s(z)) = z.$$

Thus s is injective. □

3. TWO COMBINATORIAL LEMMAS

The next two lemmas will be used in the priority construction. The first guarantees finite homogeneous rows avoiding any prescribed finite set of restraints, while the second shows that a partial ascending self-reduction of a nonrecursive non-cylinder set must have infinitely many points outside its domain.

A uniformly c.e. family $(V_x)_{x \in \mathbb{N}}$ is called *A-homogeneous* if

$$z \in V_x \implies A(z) = A(x).$$

Lemma 2 (Finite-marker avoidance). *Assume that A is not a cylinder and that neither A nor $\bar{A}$ is c.e. If $(V_x)_{x \in \mathbb{N}}$ is a uniformly c.e. A -homogeneous family, then for every finite $Z \subseteq \mathbb{N}$ there is x such that*

$$V_x \text{ is finite} \quad \text{and} \quad V_x \cap Z = \emptyset.$$

Proof. Fix a finite set $Z \subseteq \mathbb{N}$, and suppose, toward a contradiction, that there is no x such that

$$V_x \text{ is finite} \quad \text{and} \quad V_x \cap Z = \emptyset.$$

Equivalently,

$$V_x \cap Z = \emptyset \implies V_x \text{ is infinite}$$

for every x .

If $Z = \emptyset$, then every V_x is infinite. Taking $W_x = V_x$ and applying the greedy construction below gives $\text{Cyl}(A) \leq_1 A$. Since

$$x \mapsto \langle x, 0 \rangle$$

is a one-one reduction from A to $\text{Cyl}(A)$, it follows that $A \equiv_1 \text{Cyl}(A)$, contradicting the assumption that A is not a cylinder. Thus we may assume that $Z \neq \emptyset$.

For $z \in Z$, let

$$X_z = \{x : z \in V_x\}, \quad D_Z = \bigcup_{z \in Z} X_z.$$

Each X_z is c.e., and homogeneity gives

$$X_z \subseteq \begin{cases} A, & z \in A, \\ \bar{A}, & z \notin A. \end{cases}$$

Every $x \notin D_Z$ has $V_x \cap Z = \emptyset$, so V_x is infinite.

Suppose first that, for every $z \in Z$, there is some $d_z \notin D_Z$ such that

$$A(d_z) = A(z).$$

Since Z is finite, the finitely many parameters d_z may be hardcoded.

Fix a uniformly computable increasing approximation

$$V_{x,0} \subseteq V_{x,1} \subseteq \cdots, \quad V_x = \bigcup_{s \in \mathbb{N}} V_{x,s},$$

to the uniformly c.e. family $(V_x)_{x \in \mathbb{N}}$. Define

$$W_{x,s} = V_{x,s} \cup \bigcup_{\substack{z \in Z \\ z \in V_{x,s}}} V_{d_z, s}, \quad W_x = \bigcup_{s \in \mathbb{N}} W_{x,s}.$$

Thus

$$W_x = V_x \cup \bigcup_{\substack{z \in Z \\ z \in V_x}} V_{d_z},$$

and the family $(W_x)_{x \in \mathbb{N}}$ is uniformly c.e.

The family is also A -homogeneous. If $u \in V_x$, then

$$A(u) = A(x).$$

If instead

$$u \in V_{d_z} \quad \text{for some } z \in Z \cap V_x,$$

then

$$A(u) = A(d_z) = A(z) = A(x).$$

Finally, every W_x is infinite. If

$$V_x \cap Z = \emptyset,$$

then V_x is infinite by our contradictory assumption, and hence so is W_x . Otherwise, choose some

$$z \in V_x \cap Z.$$

Since $d_z \notin D_Z$, for every $w \in Z$ we have

$$d_z \notin X_w.$$

By the definition of X_w , this means that

$$w \notin V_{d_z} \quad \text{for every } w \in Z,$$

and therefore

$$V_{d_z} \cap Z = \emptyset.$$

Our contradictory assumption now implies that V_{d_z} is infinite. Since

$$V_{d_z} \subseteq W_x,$$

the set W_x is infinite as well.

Greedy construction. Fix a computable enumeration without repetitions

$$(x_0, n_0), (x_1, n_1), \dots$$

of $\mathbb{N}^2$. We construct a function

$$G : \mathbb{N}^2 \rightarrow \mathbb{N}$$

recursively along this enumeration. When treating (x_i, n_i) , wait for an element of W_{x_i} which has not been used at any earlier step, and define $G(x_i, n_i)$ to be the first such element found.

Since W_{x_i} is infinite and only finitely many values have been used before step i , this search terminates. Thus G is a total computable injection satisfying

$$G(x, n) \in W_x$$

for all x, n . By the A -homogeneity of $(W_x)_{x \in \mathbb{N}}$,

$$A(G(x, n)) = A(x).$$

Thus G witnesses

$$\text{Cyl}(A) \leq_1 A.$$

On the other hand, the map

$$x \mapsto \langle x, 0 \rangle$$

is a one-one reduction from A to $\text{Cyl}(A)$. Consequently

$$A \equiv_1 \text{Cyl}(A),$$

so A is a cylinder, contradicting our hypothesis.

Therefore, for some $z \in Z$, every number having the same A -colour as z belongs to D_Z . Let $c = A(z)$. Since Z is finite, the finite set

$$\{w \in Z : A(w) = c\}$$

may be hardcoded.

Then the entire colour class is exactly

$$\bigcup_{\substack{w \in Z \\ A(w) = c}} X_w.$$

Indeed, if $x \in X_w$ for some $w \in Z$ with $A(w) = c$, then homogeneity gives $A(x) = A(w) = c$. Conversely, suppose that $A(x) = c$. By the choice of z , we have $x \in D_Z$, so $x \in X_w$ for some $w \in Z$. Since $w \in V_x$, homogeneity gives $A(w) = A(x) = c$. Hence x belongs to the displayed union.

The displayed union is a finite union of c.e. sets. Hence either A or $\bar{A}$ is c.e., again a contradiction. $\square$

Definition 3. A partial computable function φ is a *partial ascending self-reduction of A* if, for every $x \in \text{dom } \varphi$,

$$\varphi(x) > x \quad \text{and} \quad A(\varphi(x)) = A(x).$$

Lemma 4 (Cofinite-domain lemma). *Assume that A is nonrecursive and is not a cylinder. If φ is a partial ascending self-reduction of A , then*

$$\mathbb{N} \setminus \text{dom } \varphi$$

is infinite.

Proof. Suppose that $F = \mathbb{N} \setminus \text{dom } \varphi$ is finite. Since A is nonrecursive, both A and $\bar{A}$ are infinite. For each $x \in F$, choose $t_x > x$ with $A(t_x) = A(x)$. Since F is finite, both F and the finite table $x \mapsto t_x$ may be hardcoded. Define

$$g(x) = \begin{cases} t_x, & x \in F, \\ \varphi(x), & x \notin F. \end{cases}$$

Since F is finite, membership in F is decidable after hardcoding its elements. Moreover, $\varphi(x)$ converges whenever $x \notin F$. Hence g is total computable and

$$g(x) > x, \quad A(g(x)) = A(x)$$

for every x .

Fix a computable enumeration without repetitions

$$(x_0, n_0), (x_1, n_1), \dots$$

of $\mathbb{N}^2$. We define G recursively along this enumeration. Suppose that $G(x_j, n_j)$ has already been defined for every $j < i$, and let

$$U_i = \{G(x_j, n_j) : j < i\}.$$

The set U_i is finite. Since

$$x_i < g(x_i) < g^2(x_i) < \dots,$$

the orbit $\{g^k(x_i) : k \in \mathbb{N}\}$ is infinite. Hence there is some k such that

$$g^k(x_i) \notin U_i.$$

Let

$$k_i = \mu k [g^k(x_i) \notin U_i]$$

and define

$$G(x_i, n_i) = g^{k_i}(x_i).$$

The search defining k_i is effective because g is total computable, and it terminates because the orbit of x_i is infinite whereas U_i is finite. Thus $G : \mathbb{N}^2 \rightarrow \mathbb{N}$ is total computable. It is injective by construction. Furthermore, an induction on k gives

$$A(g^k(x)) = A(x)$$

for all x, k , and hence

$$A(G(x, n)) = A(x).$$

Therefore the function

$$\langle x, n \rangle \mapsto G(x, n)$$

witnesses

$$\text{Cyl}(A) \leq_1 A.$$

Since

$$x \mapsto \langle x, 0 \rangle$$

witnesses $A \leq_1 \text{Cyl}(A)$, it follows that

$$A \equiv_1 \text{Cyl}(A).$$

Thus A is a cylinder, a contradiction. $\square$

4. THE PRIORITY CONSTRUCTION

The local interpolation theorem is proved by a priority construction relative to $0'$. We first build a Σ_2^0 set N whose complement will later be realized as the infinite-fibre profile of a computable surjection. The two lemmas of the preceding section provide the avoidance and infinitude properties needed to meet the two families of priority requirements.

Theorem 5 (Local interpolation). *Assume that A is nonrecursive, is not a cylinder, and neither A nor $\bar{A}$ is c.e. Then there is a set $C \equiv_m A$ such that*

$$[A]_{\text{fo}} < [C]_{\text{fo}} < [\text{Cyl}(A)]_{\text{fo}}.$$

Proof. Fix an effective enumeration

$$((V_x^e)_{x \in \mathbb{N}})_{e \in \mathbb{N}}$$

of all uniformly c.e. families, and an effective enumeration $(\varphi_e)_{e \in \mathbb{N}}$ of all partial computable functions.

Fix standard finite stage approximations

$$V_{x,0}^e \subseteq V_{x,1}^e \subseteq \dots, \quad V_x^e = \bigcup_s V_{x,s}^e,$$

uniformly computable in e, x, s .

We construct a set N which is c.e. in $0'$, satisfying the requirements

$$\mathcal{U}_e : (V_x^e)_x \text{ is } A\text{-homogeneous} \implies \exists x [V_x^e \text{ finite and } V_x^e \subseteq N],$$

and

$$\mathcal{L}_e : \varphi_e \text{ is a partial ascending self-reduction of } A \implies \exists m [m \notin N \cup \text{dom } \varphi_e].$$

Use the priority ordering

$$\mathcal{U}_0 > \mathcal{L}_0 > \mathcal{U}_1 > \mathcal{L}_1 > \dots.$$

The construction does not attempt to decide whether either antecedent in the requirements holds. Its actions are independent of A . The hypotheses involving A are used only in the verification: Lemma 2 is used to verify the $\mathcal{U}_e$ -requirements, while Lemma 4 is used to verify the $\mathcal{L}_e$ -requirements.

At stage s , only the first s requirements in this ordering are visited, in priority order.

Each $\mathcal{L}_e$ may carry one marker m_e . Initially, $N = \emptyset$ and no requirement carries a marker.

At every point of the construction we maintain the invariant that all active markers are pairwise distinct and lie outside the current content of N . All enumerations into N and all cancellations take effect immediately, before the next requirement is visited.

The intended roles of the requirements are as follows. After $P = \mathbb{N} \setminus N$, the map p , and $C = p^{-1}(A)$ have been defined, the $\mathcal{U}_e$ requirements will rule out finite-one reductions from $\text{Cyl}(A)$ to C , while the $\mathcal{L}_e$ requirements will rule out finite-one reductions from C to A .

Strategy for $\mathcal{U}_e$. Let Z_e be the finite set of markers currently carried by higher-priority $\mathcal{L}$ -requirements. If $\mathcal{U}_e$ has not yet acted, search among pairs $x, t \leq s$ for one such that

$$V_{x,t}^e \cap Z_e = \emptyset, \quad V_{x,t}^e \subseteq \{0, \dots, s\},$$

and

$$\forall r \geq t [V_{x,r}^e = V_{x,t}^e].$$

The last condition is decidable in $0'$. Indeed, for fixed e, x, t , failure of the stability condition is the Σ_1^0 statement

$$\exists r \geq t [V_{x,r}^e \neq V_{x,t}^e].$$

Hence $0'$ decides whether the row receives any new element after the specified stage t . Notice that this does not amount to deciding, with $0'$, whether an arbitrary c.e. row is finite.

If such a pair exists, choose the least one in a fixed computable ordering and let

$$F = V_{x,t}^e.$$

By the stability condition, $F = V_x^e$.

First cancel every active marker m_i carried by a lower-priority $\mathcal{L}_i$ such that

$$m_i \in F.$$

Then enumerate every element of F into N . Since

$$F \cap Z_e = \emptyset,$$

no higher-priority marker belongs to F , while every lower-priority marker belonging to F has already been cancelled. Hence, after the enumeration, every marker which remains active lies outside the current content of N . Declare $\mathcal{U}_e$ permanently satisfied.

If no such pair exists, do nothing.

Strategy for $\mathcal{L}_e$. If $\mathcal{L}_e$ currently has no marker, search among $m \leq s$ for an m such that

$$m \notin \text{dom } \varphi_e, \quad m \notin N_s,$$

and m is distinct from all currently active markers, where N_s denotes the current finite content of N at the moment when $\mathcal{L}_e$ is visited. The condition $m \notin \text{dom } \varphi_e$ is decidable in $0'$.

If such numbers exist, choose the least one and designate it as m_e ; otherwise do nothing. If a higher-priority $\mathcal{U}$ -requirement later selects a row containing m_e , cancel m_e before that row is enumerated into N . Every lower-priority $\mathcal{U}$ -requirement which has not yet acted includes m_e among its restraints. We now verify that all requirements are satisfied.

Verification. By definition, once a requirement $\mathcal{U}_i$ acts, it is declared permanently satisfied and is never eligible to act again. Indeed, its chosen stabilized row is then entirely contained in N . Thus every $\mathcal{U}_i$ acts at most once. Hence, for every e , there is a stage after which no $\mathcal{U}_i$ of higher priority than $\mathcal{L}_e$ acts again. From that stage on, $\mathcal{L}_e$ can never again be injured. If it already carries a marker, that marker is final. Otherwise, either it chooses a marker at some later visit, in which case that marker is final, or it never chooses another marker and remains permanently markerless. Thus every $\mathcal{L}_e$ eventually reaches a final state. Consequently, for every fixed requirement, the finitely many higher-priority marker restraints eventually stabilize.

Suppose first that φ_e is a partial ascending self-reduction. By Lemma 4,

$$D_e = \mathbb{N} \setminus \text{dom } \varphi_e$$

is infinite. Choose a stage s_0 after all higher-priority $\mathcal{U}$ -requirements have ceased acting and all higher-priority marker restraints have reached their final values.

Choose

$$m \in D_e$$

so large that $m > s_0$, that m is larger than every final higher-priority marker, and that $\mathcal{L}_e$ is among the requirements visited at stage m . Consider stage m .

Before $\mathcal{L}_e$ is visited at stage m , we have $m \notin N$. Indeed, an action performed at an earlier stage $r < m$ enumerates only numbers at most r , and therefore cannot enumerate m . At stage m , before $\mathcal{L}_e$ is visited, only higher-priority requirements have been visited. The higher-priority $\mathcal{L}$ -requirements do not enumerate into N , and no higher-priority $\mathcal{U}$ -requirement acts after s_0 . Hence no action before the visit of $\mathcal{L}_e$ at stage m can have placed m into N .

Moreover, m is not an active marker when $\mathcal{L}_e$ is visited. All higher-priority markers have already reached their final values and are smaller than m , while a lower-priority marker chosen at an earlier stage $r < m$ is also smaller than m , since markers chosen at stage r are at most r . Lower-priority requirements have not yet been visited at stage m .

Any marker already carried by $\mathcal{L}_e$ itself was also chosen at an earlier stage $r < m$, and hence is smaller than m .

If $\mathcal{L}_e$ already carries a marker when it is visited at stage m , let m_e denote this marker. By construction,

$$m_e \notin \text{dom } \varphi_e.$$

No higher-priority $\mathcal{U}$ -requirement acts after s_0 . Every lower-priority $\mathcal{U}$ -requirement which had not yet acted when m_e was chosen includes m_e among its restraints and hence, if it acts later, selects a row disjoint from $\{m_e\}$.

On the other hand, if a lower-priority $\mathcal{U}$ -requirement had already acted before m_e was chosen, its selected row was already contained in N at the time of the choice. Since m_e was chosen outside the current content of N , that row cannot contain m_e . Therefore

$$m_e \notin N \cup \text{dom } \varphi_e,$$

and $\mathcal{L}_e$ is satisfied.

Otherwise, $\mathcal{L}_e$ has no marker when it is visited at stage m . By the choice of m and the preceding observations, m itself is outside N , outside $\text{dom } \varphi_e$, and distinct from all active markers. Thus m is a candidate in the bounded search performed by $\mathcal{L}_e$, and hence the strategy chooses some marker m_e .

No higher-priority $\mathcal{U}$ -requirement acts thereafter, and every lower-priority $\mathcal{U}$ -requirement which has not yet acted treats m_e as a restraint. If a lower-priority $\mathcal{U}$ -requirement acted before m_e was chosen, its selected row is already contained in N , whereas m_e was chosen outside the current content of N . Hence that row cannot contain m_e . Therefore

$$m_e \notin N \cup \text{dom } \varphi_e.$$

If a higher-priority marker is later defined or redefined, its new value is chosen outside the current content of N . Since every row previously selected by a $\mathcal{U}$ -strategy is already contained in N , the new marker cannot belong to any such row. Thus a previous $\mathcal{U}$ -action remains compatible with every later definition or redefinition of a higher-priority marker.

Now suppose that $(V_x^e)_x$ is A -homogeneous. If $\mathcal{U}_e$ has already acted, it is permanently satisfied. Otherwise, after all higher-priority marker restraints have reached their final values, let Z be the resulting finite set of markers. At every subsequent visit of $\mathcal{U}_e$, the set Z_e used by the strategy is therefore equal to Z .

By Lemma 2, there is an x such that

$$V_x^e \text{ is finite} \quad \text{and} \quad V_x^e \cap Z = \emptyset.$$

Since V_x^e is finite, choose a stage t by which every element of V_x^e has been enumerated and after which no new element enters the row. Thus

$$V_{x,t}^e = V_x^e$$

and

$$V_{x,r}^e = V_{x,t}^e \quad \text{for every } r \geq t.$$

For every sufficiently large stage s at which $\mathcal{U}_e$ is visited, we have

$$x, t \leq s, \quad V_{x,t}^e \cap Z_e = \emptyset, \quad V_{x,t}^e \subseteq \{0, \dots, s\}.$$

Thus (x, t) satisfies all the tests in the $\mathcal{U}_e$ -strategy. Since every fixed requirement is visited at all sufficiently large stages, the bounded $0'$ -search eventually makes $\mathcal{U}_e$ act. Therefore every $\mathcal{U}_e$ is satisfied.

At each stage only finitely many requirements are visited, each strategy performs only finitely many $0'$ -decidable tests, and only a finite set is enumerated into N . Thus the enumeration of N is computable in $0'$.

Let

$$P = \mathbb{N} \setminus N.$$

Then $P \in \Pi_2^0$. Although N was enumerated relative to $0'$, the profile-realization lemma converts the resulting Π_2^0 set P into the infinite-fibre profile of an ordinary computable function. By Lemma 1, fix a total computable surjection p with computable section such that

$$I_p = P,$$

and put

$$C = p^{-1}(A).$$

Since

$$I_p = P = \mathbb{N} \setminus N,$$

we have

$$y \in P \iff p^{-1}(y) \text{ is infinite,}$$

and

$$y \in N \iff p^{-1}(y) \text{ is finite.}$$

This is the link between the priority requirements and the finite-one degree of C : the $\mathcal{U}_e$ requirements exploit finite fibres over N , whereas the $\mathcal{L}_e$ requirements exploit infinite fibres over P . The section gives $A \leq_1 C$, while p gives $C \leq_m A$, so $C \equiv_m A$.

We first show that

$$C <_{\text{fo}} \text{Cyl}(A).$$

There is a one-one reduction

$$u \mapsto \langle p(u), u \rangle$$

from C to $\text{Cyl}(A)$. Suppose conversely that

$$h : \text{Cyl}(A) \leq_{\text{fo}} C.$$

For each x , define

$$V_x = \{p(h(\langle x, n \rangle)) : n \in \mathbb{N}\}.$$

Although p was chosen only after the construction of N , there is no circularity here. Once p and the hypothetical reduction h are fixed, $(V_x)_{x \in \mathbb{N}}$ is an ordinary uniformly c.e. family. Consequently, there is an index e such that

$$V_x = V_x^e \quad \text{for every } x.$$

This family is A -homogeneous. Indeed, if

$$y = p(h(\langle x, n \rangle)),$$

then

$$y \in A \iff h(\langle x, n \rangle) \in C \iff \langle x, n \rangle \in \text{Cyl}(A) \iff x \in A.$$

Hence requirement $\mathcal{U}_e$, for the index e fixed above, supplies an x for which $V_x = V_x^e$ is finite and

$$V_x \subseteq N = \mathbb{N} \setminus P.$$

For every $y \in V_x$, the fibre $p^{-1}(y)$ is finite. Therefore

$$h(\{\langle x, n \rangle : n \in \mathbb{N}\}) \subseteq \bigcup_{y \in V_x} p^{-1}(y),$$

a finite set. This is impossible, since a finite-one function cannot map an infinite set into a finite set.

Thus

$$C <_{\text{fo}} \text{Cyl}(A).$$

It remains to show that

$$A <_{\text{fo}} C.$$

Suppose toward a contradiction that

$$h : C \leq_{\text{fo}} A.$$

Define the partial computable function f_h by

$$f_h(y) = h(u),$$

where u is the first number found such that

$$p(u) = y \quad \text{and} \quad h(u) > y.$$

Whenever $f_h(y)$ is defined,

$$f_h(y) > y$$

and, because h is a reduction,

$$A(f_h(y)) = A(h(u)) = A(p(u)) = A(y).$$

Thus f_h is a partial ascending self-reduction of A .

If $y \in P$, then $p^{-1}(y)$ is infinite. Since h is finite-one,

$$h^{-1}(\{0, \dots, y\}) = \bigcup_{z \leq y} h^{-1}(z)$$

is finite. Hence some $u \in p^{-1}(y)$ satisfies $h(u) > y$, and so

$$P \subseteq \text{dom } f_h.$$

Again, there is no circularity in applying one of the requirements fixed before the construction. Once p and the hypothetical reduction h are fixed, f_h is an ordinary partial computable function. Hence there is an index e such that

$$\varphi_e = f_h.$$

By the verification of $\mathcal{L}_e$, there is a final marker m_e such that

$$m_e \notin N \quad \text{and} \quad m_e \notin \text{dom } f_h.$$

But $m_e \notin N$ means $m_e \in P$, contradicting $P \subseteq \text{dom } f_h$. Hence $C \not\leq_{\text{fo}} A$ and therefore

$$A <_{\text{fo}} C.$$

This proves the theorem. □

Remark 6. Two features of the construction are crucial.

First, for a partial computable function φ_e , the condition

$$m \notin \text{dom } \varphi_e$$

is decidable in $0'$. Thus a marker can be chosen with permanent certification that it lies outside $\text{dom } \varphi_e$.

Second, a $\mathcal{U}$ -requirement acting at stage s enumerates into N only numbers at most s .

Consequently, an action performed at a stage $s < m$ cannot enumerate m into N ; hence no action before stage m can do so.

After the higher-priority $\mathcal{U}$ -requirements have ceased acting, a newly chosen marker is protected from all lower-priority strategies. A lower-priority $\mathcal{U}$ -requirement which has already acted selected a row already contained in N , whereas the marker is chosen outside the current content of N ; hence that row cannot contain the marker. Every lower-priority $\mathcal{U}$ -requirement which has not yet acted includes the marker among its restraints and therefore selects a row disjoint from it. The finite parameters hardcoded in Lemmas 2 and 4 do not enter the priority construction. Those lemmas are used only existentially in the verification. For a $\mathcal{U}_e$ -requirement, the current finite restraint set Z_e is explicitly available at every stage; once it stabilizes, Lemma 2 guarantees the existence of a finite row disjoint from it, and the bounded $0'$ -search finds a stabilization pair (x, t) without having to recognize that the restraints have stabilized. For a $\mathcal{L}_e$ -requirement, Lemma 4 is used only to infer that $\mathbb{N} \setminus \text{dom } \varphi_e$ is infinite, while each individual

condition $m \notin \text{dom } \varphi_e$ is decided directly by $0'$. Thus no uniform recovery of the finite sets or auxiliary parameters used in the proofs of the two lemmas is required.

5. THE DICHOTOMY

We now combine the local interpolation theorem with Maslova's dichotomy for nonrecursive c.e. many-one degrees to obtain the global one-or-infinity result.

We use the following consequence of Maslova's theorem [2], in the formulation recorded by Richter, Stephan, and Zhang [6, p. 17]: every nonrecursive c.e. many-one degree contains either exactly one or infinitely many finite-one degrees.

Corollary 7. *Every nonrecursive many-one degree contains either exactly one or infinitely many finite-one degrees. In particular, there is no nonrecursive many-one degree containing at least two but only finitely many finite-one degrees.*

Proof. For a many-one degree $\mathbf{d}$, let

$$\mathcal{F}(\mathbf{d}) = \{[X]_{\text{fo}} : X \in \mathbf{d}\}.$$

Suppose, toward a contradiction, that $\mathbf{d}$ is nonrecursive and that

$$1 < |\mathcal{F}(\mathbf{d})| < \aleph_0.$$

Let T be the greatest element of $\mathcal{F}(\mathbf{d})$, that is

$$T = [\text{Cyl}(X)]_{\text{fo}}$$

for every $X \in \mathbf{d}$. Since $\mathcal{F}(\mathbf{d}) \setminus \{T\}$ is finite and nonempty, there is an element $E < T$ which is covered by T ; that is, there is no $D \in \mathcal{F}(\mathbf{d})$ such that

$$E < D < T.$$

Choose a set $A \in \mathbf{d}$ such that

$$[A]_{\text{fo}} = E.$$

Since $A \in \mathbf{d}$ and $\mathbf{d}$ is nonrecursive, A is nonrecursive. Moreover, A is not a cylinder. Indeed, if A were a cylinder, then

$$E = [A]_{\text{fo}} = [\text{Cyl}(A)]_{\text{fo}} = T,$$

because $[\text{Cyl}(A)]_{\text{fo}}$ is the greatest finite-one degree inside $[A]_m = \mathbf{d}$. This contradicts $E < T$.

If A is c.e., then $\mathbf{d} = [A]_m$ is a nonrecursive c.e. many-one degree containing finitely many but more than one finite-one degrees, contrary to Maslova's theorem.

Suppose next that $\overline{A}$ is c.e. Put

$$\mathbf{d}^c = [\overline{A}]_m.$$

Complementation induces an order isomorphism

$$\Phi : \mathcal{F}(\mathbf{d}) \longrightarrow \mathcal{F}(\mathbf{d}^c)$$

given by

$$\Phi([X]_{\text{fo}}) = [\overline{X}]_{\text{fo}}.$$

Indeed, for every total computable function f ,

$$f : X \leq_m Y \iff f : \overline{X} \leq_m \overline{Y},$$

and, since the fibres of f are unchanged,

$$f : X \leq_{\text{fo}} Y \iff f : \overline{X} \leq_{\text{fo}} \overline{Y}.$$

Thus Φ is well defined and order preserving. The same complementation map, now considered from $\mathcal{F}(\mathbf{d}^c)$ to $\mathcal{F}(\mathbf{d})$, is the inverse of Φ . Hence Φ is an order isomorphism.

Consequently,

$$|\mathcal{F}(\mathbf{d}^c)| = |\mathcal{F}(\mathbf{d})|,$$

so $\mathbf{d}^c$ also contains finitely many but more than one finite-one degrees. Since $\overline{A}$ is c.e. and nonrecursive, this again contradicts Maslova's theorem.

We may therefore assume that neither A nor $\bar{A}$ is c.e. Theorem 5 now yields a set $C \equiv_m A$ such that

$$[A]_{\text{fo}} < [C]_{\text{fo}} < [\text{Cyl}(A)]_{\text{fo}}.$$

Since $C \equiv_m A$, the degree $[C]_{\text{fo}}$ belongs to $\mathcal{F}(\mathbf{d})$. Moreover,

$$[A]_{\text{fo}} = E \quad \text{and} \quad [\text{Cyl}(A)]_{\text{fo}} = T.$$

Hence

$$E < [C]_{\text{fo}} < T,$$

contradicting the fact that E is covered by T .

Therefore $\mathcal{F}(\mathbf{d})$ cannot be finite of cardinality greater than one. Since every many-one degree contains at least one finite-one degree, it follows that every nonrecursive many-one degree contains either exactly one or infinitely many finite-one degrees. $\square$

6. CONCLUSION

We have shown that every nonrecursive many-one degree contains either exactly one or infinitely many finite-one degrees, giving a negative answer to Open Question 2 of Richter–Stephan–Zhang.

The main ingredient is the local interpolation theorem: whenever A is nonrecursive, is not a cylinder, and neither A nor $\bar{A}$ is c.e., the finite-one degree of A can be strictly interpolated below the greatest finite-one degree in $[A]_m$. Together with Maslova’s theorem for c.e. many-one degrees and complementation, this local phenomenon rules out every finite nontrivial configuration of finite-one degrees inside a nonrecursive many-one degree.

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